

Personality Profiling with Clustering

1 Introduction

The objective of this assignment is to apply various unsupervised clustering algorithms to the "Big Five Personality Test" dataset to identify distinct personality profiles among individuals. We will evaluate the performance of these clustering models using appropriate metrics and visualize the resulting clusters to gain insights into human personality traits.

2 Data

2.1 Dataset

The Big Five Personality Test dataset comprises responses from individuals on a personality assessment. The features are questions related to each personality type and the values are scores given to them by participants from 1 to 5.

2.2 Features

The features are as follows which are numerical:

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['EXT1', 'EXT2', 'EXT3', 'EXT4', 'EXT5', 'EXT6', 'EXT7', 'EXT8', 'EXT9', 'EXT10', 'EST1', 'EST2', 'EST3', 'EST4', 'EST5', 'EST6', 'EST7', 'EST8', 'EST9', 'EST10', 'AGR1', 'AGR2', 'AGR3', 'AGR4', 'AGR5', 'AGR6', 'AGR7', 'AGR8', 'AGR9', 'AGR10', 'CSN1', 'CSN2', 'CSN3', 'CSN4', 'CSN5', 'CSN6', 'CSN7', 'CSN8', 'CSN9', 'CSN10', 'OPN1', 'OPN2', 'OPN3', 'OPN4', 'OPN5', 'OPN6', 'OPN7', 'OPN8', 'OPN9', 'OPN10', 'EXT1_E', 'EXT2_E', 'EXT3_E', 'EXT4_E', 'EXT5_E', 'EXT6_E', 'EXT7_E', 'EXT8_E', 'EXT9_E', 'EXT10_E', 'EST1_E', 'EST2_E', 'EST3_E', 'EST4_E', 'EST5_E', 'EST6_E', 'EST7_E', 'EST8_E', 'EST9_E', 'EST10_E', 'AGR1_E', 'AGR2_E', 'AGR3_E', 'AGR4_E', 'AGR5_E', 'AGR6_E', 'AGR7_E', 'AGR8_E', 'AGR9_E', 'AGR10_E', 'CSN1_E', 'CSN2_E', 'CSN3_E', 'CSN4_E', 'CSN5_E', 'CSN6_E', 'CSN7_E', 'CSN8_E', 'CSN9_E', 'CSN10_E', 'OPN1_E', 'OPN2_E', 'OPN3_E', 'OPN4_E', 'OPN5_E', 'OPN6_E', 'OPN7_E', 'OPN8_E', 'OPN9_E', 'OPN10_E', 'dateload', 'screenw', 'screenh', 'introelapse', 'testelapse', 'endelapse', 'IPC', 'country', 'lat_appx_lots_of_err', 'long_appx_lots_of_err']
```

The dataset contains 110 cols (features) and 1015341 rows.

Then I checked the missing values which are as follows:

Missing Values:

EXT1: 1783

EXT2: 1783

EXT3: 1783

EXT4: 1783

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OPN5_E: 1783

OPN6_E: 1783

OPN7_E: 1783

OPN8_E: 1783

OPN9_E: 1783

OPN10_E: 1783

screenw: 2066

screenh: 2066

introelapse: 2066

testelapse: 1783

Since the number of missing vals are too few in comparison to the data so we remove them all.

Since scores are expected to be integers from 1 to 5, values outside this range or extreme values can be considered outliers.

I chose only 50 features that were usefull in our task as follows:

Personality features :

'EXT1', 'EXT2', 'EXT3', 'EXT4', 'EXT5', 'EXT6', 'EXT7', 'EXT8', 'EXT9', 'EXT10',

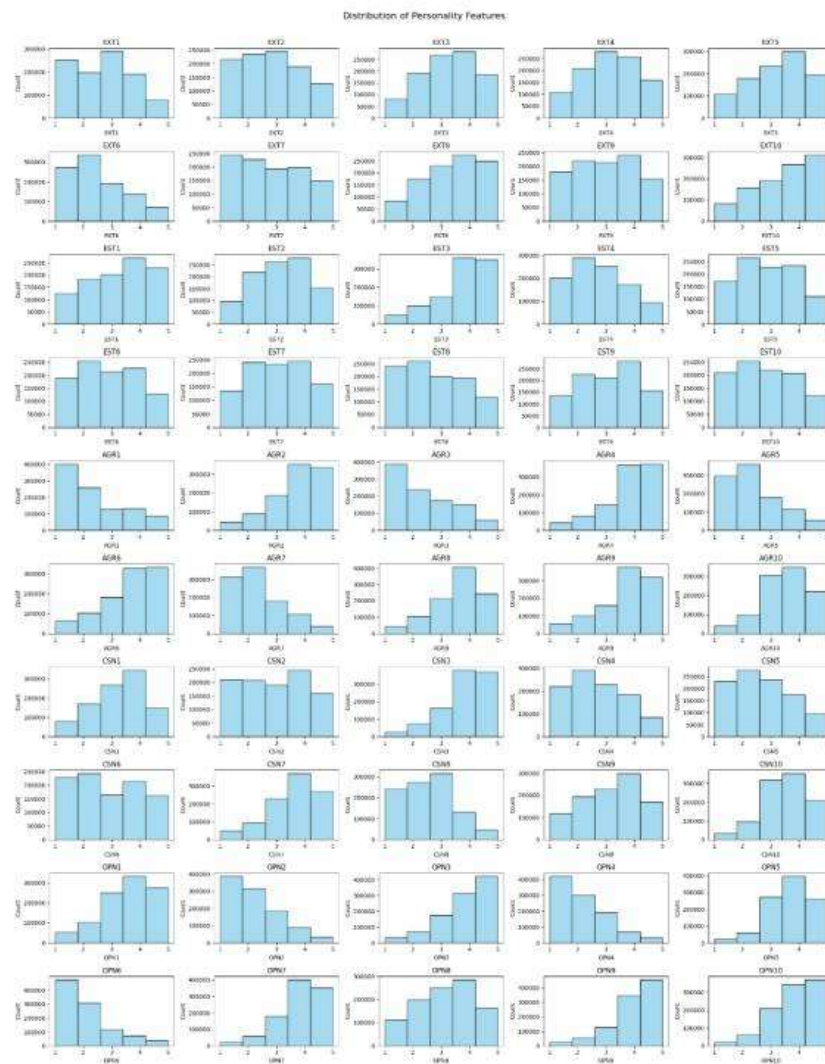
'EST1', 'EST2', 'EST3', 'EST4', 'EST5', 'EST6', 'EST7', 'EST8', 'EST9', 'EST10',

'AGR1', 'AGR2', 'AGR3', 'AGR4', 'AGR5', 'AGR6', 'AGR7', 'AGR8', 'AGR9', 'AGR10',

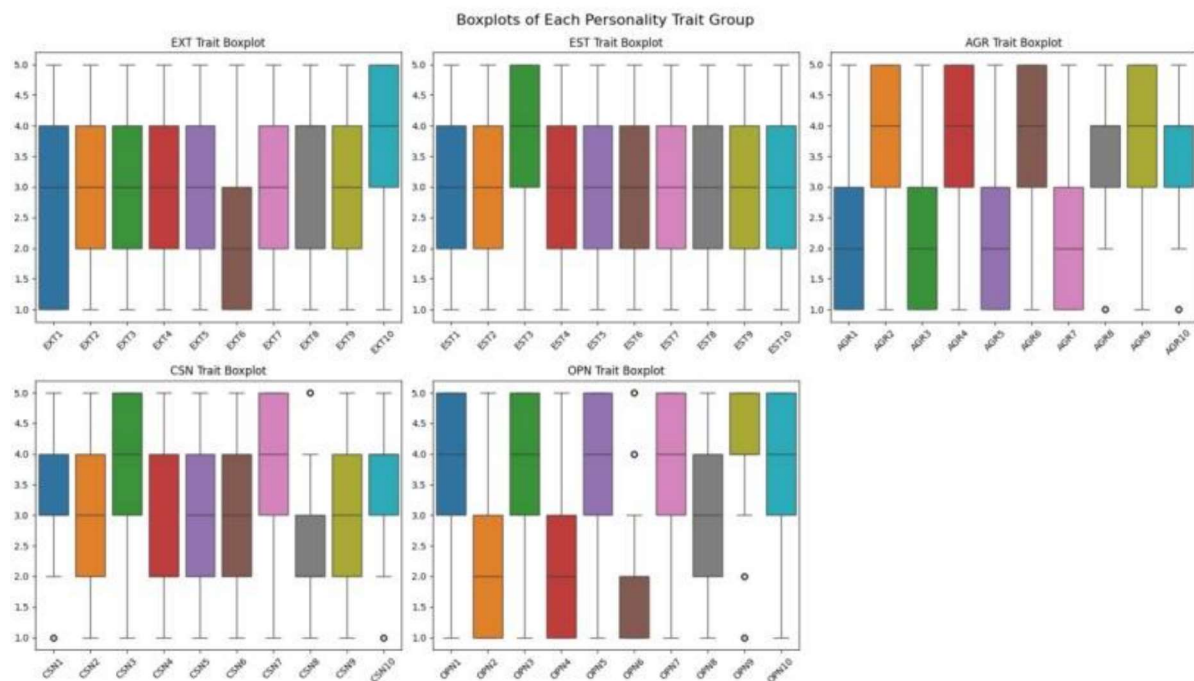
'CSN1', 'CSN2', 'CSN3', 'CSN4', 'CSN5', 'CSN6', 'CSN7', 'CSN8', 'CSN9', 'CSN10',
'OPN1', 'OPN2', 'OPN3', 'OPN4', 'OPN5', 'OPN6', 'OPN7', 'OPN8', 'OPN9', 'OPN10',

2.3 visualization

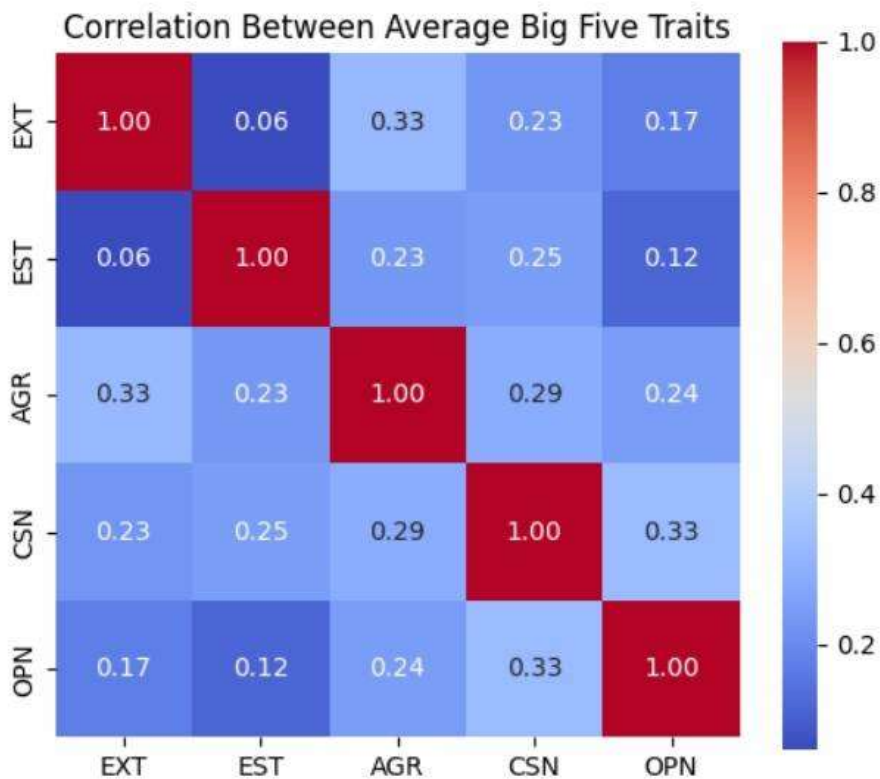
In this section we will go through some visualization related to the data that can help us to have a better sense of our data and the relationship between the features.



In this plot we can see the distribution of the scores through each feature and personality.

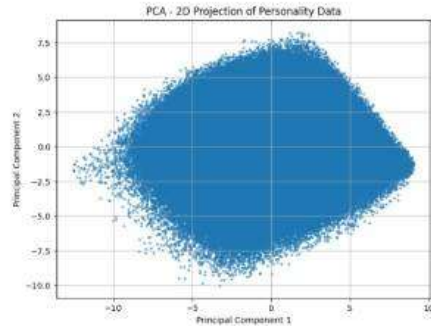


The image display boxplots for 5 major personality trait groups: EXT, EST , AGR , CSN and OPN. The boxplots shows the distribution and spread of scores within each trait allow for comparison across the different personality dimensions. The plots reveal differences in the central tendencies and variability of the traits providing insights into the underlying personality characteristics of the sample. This type of visualization can be useful for understanding individual differences and patterns in personality profiles.



The image shows a correlation matrix between the five major personality traits. The correlation between Extraversion (EXT) and Agreeableness (AGR) is 0.33, showing a moderate positive relationship between these two traits. The correlation between Conscientiousness (CSN) and Openness (OPN) is 0.33, also suggesting a moderate positive relationship. This correlation matrix provides insights into the relationships between the Big Five personality traits and can be useful for understanding individual differences and patterns in personality profiles.

3 Dimensionality Reduction and Scaling



Here We performed PCA to perform dimintinality reduction to visualize the data in lower dimensions before clustering, to get a sense of inherent groupings.

I performed StandardScaler which Transforms features to have zero mean and unit variance. After scaling all features contribute equally to the distance metrics. we Use StandardScaler here because personality scores are likely centered around a mid-point (e.g., 3), we're using clustering algorithms like KMeans, DBSCAN, Hierarchical Clustering, etc.

Why is Scaling Important? Clustering algorithms (especially distance based ones) rely on distance metrics like Euclidean distance. If the features are not on the same scale Features with larger ranges will dominate the distance calculations, leading to biased clustering. This can result in misleading clusters where the model gives undue importance to some variables.

4 Clustering Model Development and Evaluation

K-means:

Here we use a algorithm to find the best k according to metrics like Silhouette and Davies-Bouldin as well as elbow method. The results for each k is as follows:

k=2 | Inertia=451913.01 | Silhouette=0.0893 | Davies-Bouldin=3.0001

k=3 | Inertia=430094.84 | Silhouette=0.0718 | Davies-Bouldin=3.0745

k=4 | Inertia=414948.02 | Silhouette=0.0618 | Davies-Bouldin=3.0784

k=5 | Inertia=404313.56 | Silhouette=0.0518 | Davies-Bouldin=2.9764

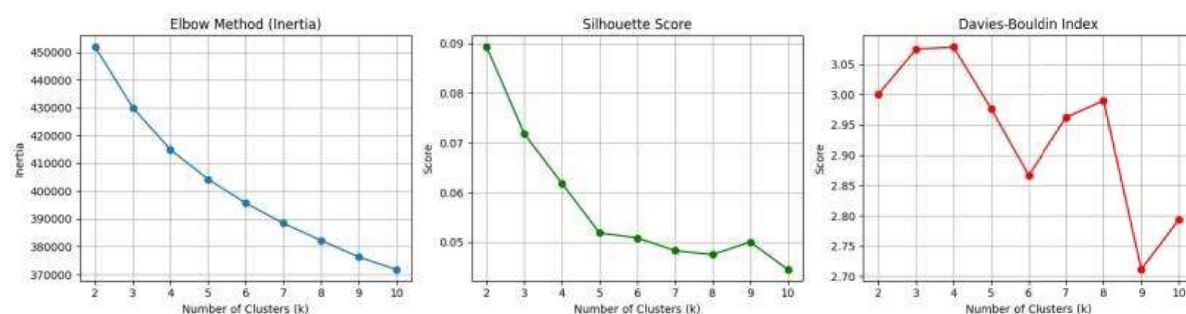
k=6 | Inertia=395671.58 | Silhouette=0.0509 | Davies-Bouldin=2.8666

k=7 | Inertia=388426.02 | Silhouette=0.0483 | Davies-Bouldin=2.9617

k=8 | Inertia=382255.13 | Silhouette=0.0475 | Davies-Bouldin=2.9897

k=9 | Inertia=376304.80 | Silhouette=0.0500 | Davies-Bouldin=2.7113

k=10 | Inertia=371775.98 | Silhouette=0.0445 | Davies-Bouldin=2.7939



After testing different values for k in the K-Means clustering model two specific options stand out based on the results:

Option 1: k = 2 This value gives the highest silhouette score (0.0893), which means the clusters are more distinct from one another.

It also keeps the structure simple—just two main personality groups.

If we want to looking to separate individuals into broader, more general categories, k = 2 works well.

Option 2: k = 9 This setup gives the lowest Davies-Bouldin Index (2.71), which indicates better separation between clusters and more compact groups.

While the silhouette score is lower than with k = 2, it's still reasonable (0.0500).

This value is more suitable if we want to capture a wider variety of personality types and get a more detailed view of the data.

So I choose k=9 as the best k.

Hierarchical Clustering:

In this section we try to find the best linkage method as well as the best k:

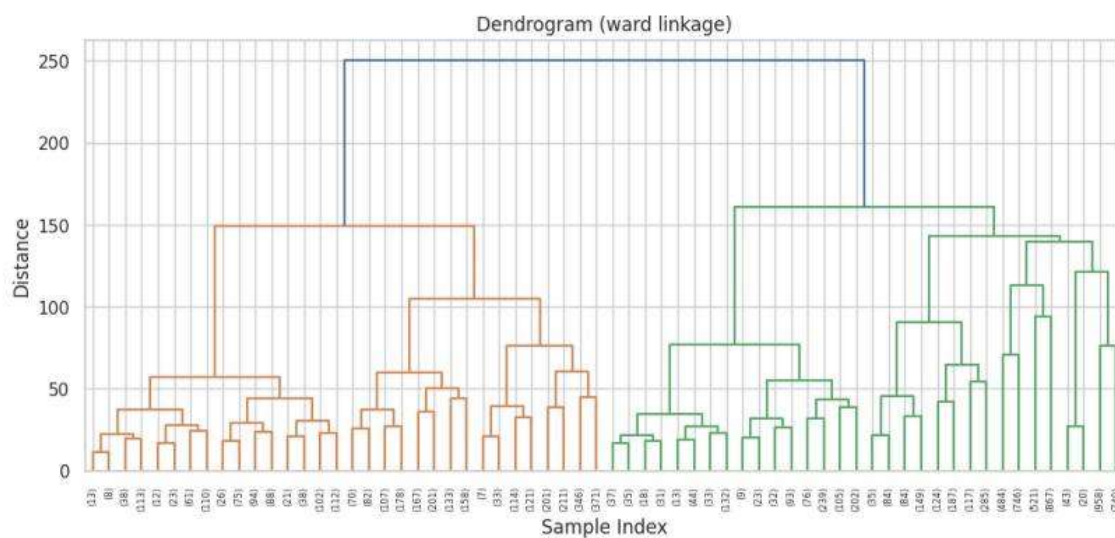
After running hierarchical clustering with different linkage methods and values for k, the results show that the average linkage method gave the best performance overall.

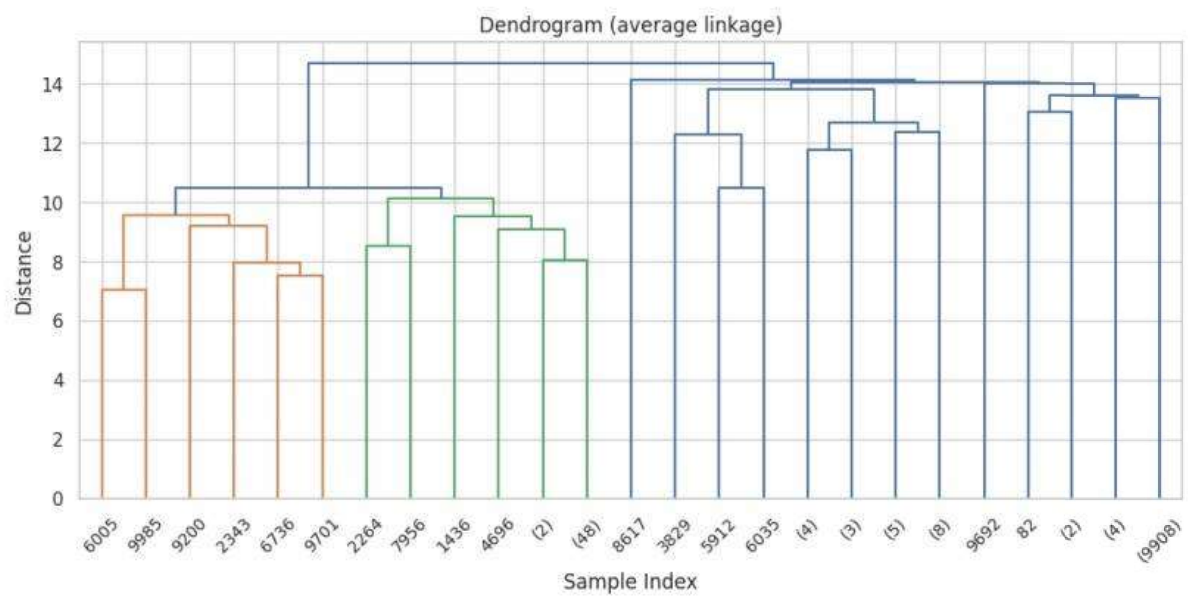
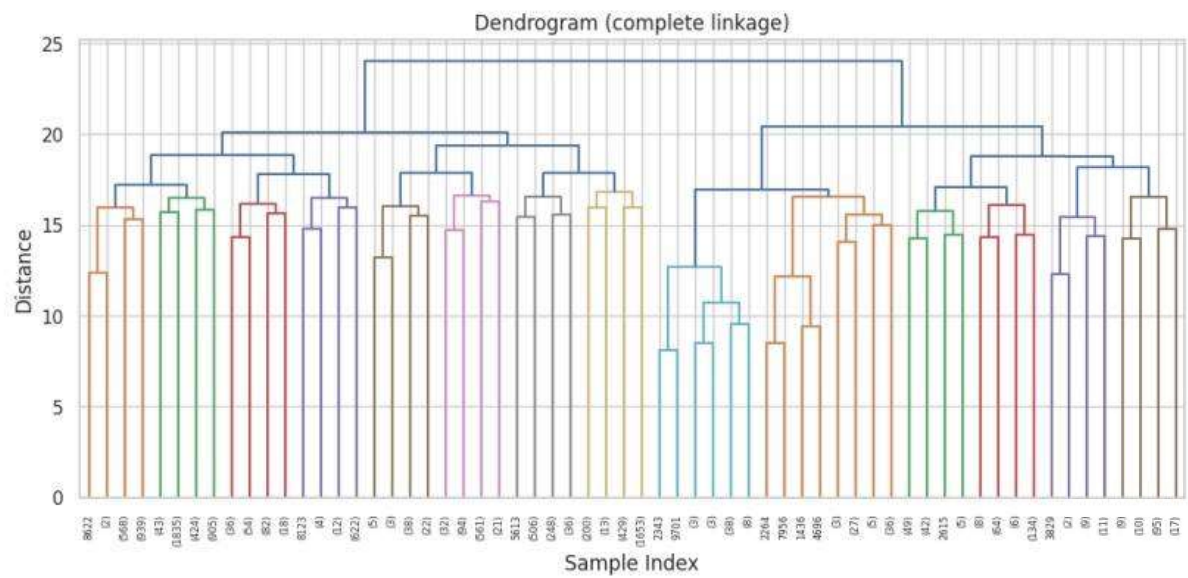
Choosing the Best k: Looking at the results:

When using average linkage with k = 2, the Silhouette Score is the highest at 0.334, and the Davies-Bouldin Index is reasonably low at 0.92. This means the clusters are fairly well-separated and internally consistent.

With k = 3, the Silhouette Score drops slightly to 0.288, but the Davies-Bouldin Index improves further to 0.79, suggesting slightly better separation between clusters. If the goal is to split the data into a few large and distinct groups, k = 2 is the better choice. It gives strong separation and is easy to interpret. On the other hand if we're looking for a more detailed breakdown of personality types and are okay with a bit more overlap between groups, k = 3 is a solid option.

So we can go with k = 2 using average linkage if we want clear and simple clusters. We can use k = 3 if we want a bit more depth and are exploring personality subtypes. I go with k=3.



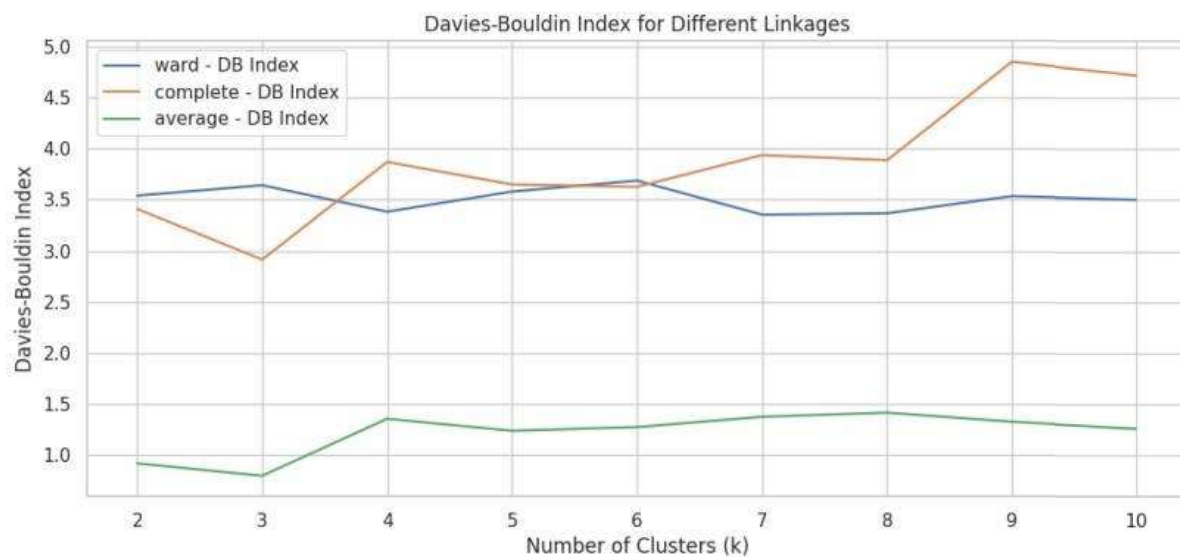
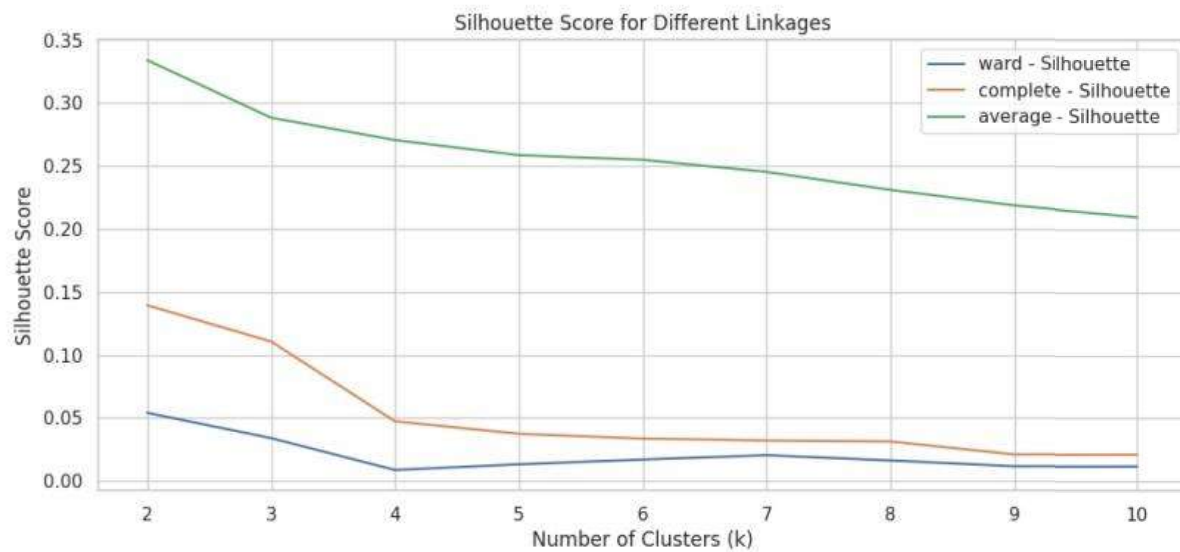


Clustering Evaluation Results:

	Linkage	k	Silhouette Score	Davies-Bouldin Index
18	average	2	0.333964	0.920599
19	average	3	0.288328	0.799035
20	average	4	0.270561	1.356797

	Linkage	k	Silhouette Score	Davies-Bouldin Index
21	average	5	0.258610	1.239424
22	average	6	0.254958	1.275627
23	average	7	0.245320	1.376276
24	average	8	0.231055	1.416791
25	average	9	0.218792	1.328005
26	average	10	0.209709	1.262202
9	complete	2	0.139448	3.409029
10	complete	3	0.110709	2.913919
11	complete	4	0.047404	3.868836
12	complete	5	0.037490	3.648263
13	complete	6	0.033691	3.626090
14	complete	7	0.032179	3.936587
15	complete	8	0.031409	3.886016
17	complete	10	0.021283	4.719060
16	complete	9	0.021108	4.849503
0	ward	2	0.054156	3.539427
1	ward	3	0.034049	3.641383
5	ward	7	0.020539	3.353763
4	ward	6	0.017078	3.686235
6	ward	8	0.016327	3.366589

	Linkage	k	Silhouette Score	Davies-Bouldin Index
3	ward	5	0.013287	3.579552
8	ward	10	0.011902	3.502685
7	ward	9	0.011759	3.534929
2	ward	4	0.008785	3.382308



DBSCAN:

DBSCAN clusters points with high densities together but marks points in sparse regions as noise (outliers). DBSCAN also has two significant parameters:

eps: Radius of the neighbors to use.

min_samples: Number of points required to form a dense region.

we can experiment with changing eps and min_samples values to find an effective clustering similar to your data's pattern.

Noise Points Implications Noise points that are marked as (-1) are considered outliers or nonexistent points in any cluster.

Having lots of noise points might mean the eps is too small or data is too sparse.

Noise points may be indicating people with uncommon personality traits or measurement error.

Evaluation Metrics Silhouette Score: It calculates how close the points are to their own cluster and other clusters. It is normalized between -1 and 1; higher values are desired.

Davies-Bouldin Index: It calculates average similarity of each cluster with its best similarity match. Lower values are desirable.

We only compute these metrics if DBSCAN finds at least 2 clusters (otherwise, the scores don't hold).

Implication of Noise Points in DBSCAN Noise points in DBSCAN are data points that are noticed by the algorithm as not belonging to any cluster because they lie in the region of low density. These are labeled as -1 and are dealt with virtually like outliers or anomalies.

What does this mean for your personality data set? Outliers or Unusual Personality Profiles
Noisy points may be those people whose personality answers are radically different or inconsistent from the others. They might be people with extremely unusual or uncommon combinations of traits that do not belong to the bigger clusters.

Data Quality Issues Noise sometimes may indicate data flaws, i.e., not treated missing values, faulty questionnaires, or measurement inaccuracies.

Sensitivity of Parameters The number of noise points significantly relies on eps and min_samples parameters:

eps too low: A too small neighborhood radius marks too many points as noise and shatters clusters.

Too large eps: Can have fewer noise points but much too large clusters grouping into separate clusters.

Too large min_samples: Requires more dense areas to group, hence more noise points.

Interpretation of Clustering Quality Large number of noise points may be a sign that data naturally doesn't cluster well at these parameter values or data is very noisy or heterogeneous.DBSCAN clustering results sorted by Silhouette Score:

	eps	min_samples	n_clusters	n_noise	Silhouette Score \
0	3.1	5	3	497	0.155756
1	2.8	5	3	1134	0.137856
2	2.8	4	4	1041	0.126562
3	3.1	4	3	457	0.113862
4	3.1	3	5	428	0.044645
5	2.5	5	5	2345	-0.049026
6	2.8	3	12	934	-0.101655
7	1.0	3	2	9962	-0.120995
8	1.0	5	2	9962	-0.120995
9	1.0	4	2	9962	-0.120995
10	3.1	2	16	406	-0.123211
11	1.3	5	2	9958	-0.129107
12	2.5	4	11	2146	-0.173205
13	2.5	3	26	1927	-0.247924
14	2.8	2	53	852	-0.259349
15	2.2	5	17	4561	-0.290338
16	2.2	4	23	4256	-0.327579
17	2.5	2	129	1721	-0.364886
18	1.3	4	9	9922	-0.381886
19	2.2	3	81	3790	-0.387157
20	1.9	5	47	7586	-0.426153
21	2.2	2	317	3318	-0.428283
22	1.6	5	33	9674	-0.431358

23	1.9	4	68	7081	-0.433826
24	1.0	2	22	9922	-0.437568
25	1.9	2	513	5657	-0.442657
26	1.3	3	29	9858	-0.447931
27	1.9	3	137	6409	-0.449942
28	1.6	4	72	9415	-0.459047
29	1.6	2	528	8202	-0.461451
30	1.6	3	153	8952	-0.469952
31	1.3	2	174	9568	-0.511839

Davies-Bouldin Index

0	2.653096
1	2.893074
2	2.644110
3	2.610792
4	2.209143
5	2.826353
6	1.788448
7	1.069352
8	1.069352
9	1.069352
10	1.652641
11	1.172332
12	2.158333
13	1.657811
14	1.377369
15	2.004701

16	1.849990
17	1.330877
18	2.461819
19	1.532339
20	1.968782
21	1.343535
22	2.430787
23	1.717489
24	1.571095
25	1.428915
26	2.133108
27	1.640877
28	2.065965
29	1.613726
30	1.877167
31	1.688323

Best DBSCAN parameters found:

eps = 3.1000000000000005

min_samples = 5.0

Number of clusters: 3.0

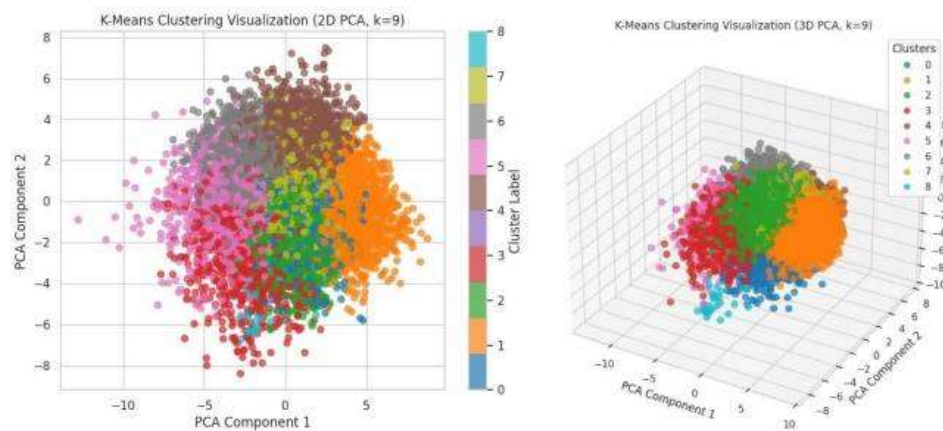
Number of noise points: 497.0

Silhouette Score: 0.1558

Davies-Bouldin Index: 2.6531

5 Interpretation & Visualization

K-means:



Trait1 Trait2 Trait3 Trait4 Trait5 Trait6 Trait7 \

Cluster

0	0.608283	-0.495513	0.526269	-0.595006	0.480000	-0.351991	0.536906
1	0.696885	-0.721840	0.926904	-0.833152	0.896328	-0.711912	0.859566
2	-0.517280	0.606399	-0.146662	0.564357	-0.430084	0.308579	-0.520754
3	-0.651856	0.670981	-0.826684	0.584406	-0.806961	0.379252	-0.763062
4	0.620615	-0.702776	0.380206	-0.578785	0.596972	-0.589451	0.608588
5	-0.488982	0.542226	-0.601695	0.468115	-0.688555	0.833088	-0.574807
6	-0.675090	0.555417	-0.839639	0.798496	-0.691789	0.358879	-0.741814
7	0.346279	-0.289896	0.392785	-0.231131	0.384144	-0.048478	0.352608
8	-0.569964	-0.930184	-1.473170	-1.568466	-1.480523	-1.076116	-1.176539

Trait8 Trait9 Trait10 ... Trait41 Trait42 Trait43 \

Cluster

0	-0.605504	0.589153	-0.536237	...	0.002438	-0.080178	-0.139707
1	-0.523545	0.635028	-0.925620	...	0.362291	-0.484035	0.228742
2	0.500209	-0.495487	0.457931	...	0.094689	-0.299951	0.042234
3	0.367622	-0.330946	0.579510	...	0.385871	-0.386947	0.172773
4	-0.539938	0.638852	-0.434884	...	0.298173	-0.210143	0.438804

5	0.339077	-0.450553	0.464467	...	-0.818869	0.964044	-0.708143
6	0.594463	-0.630180	0.712153	...	0.204401	-0.054350	0.388935
7	-0.119940	0.116490	-0.213450	...	-0.485499	0.571600	-0.446095
8	-1.821110	-1.280897	-1.756969	...	-1.904210	-0.638025	-2.344659

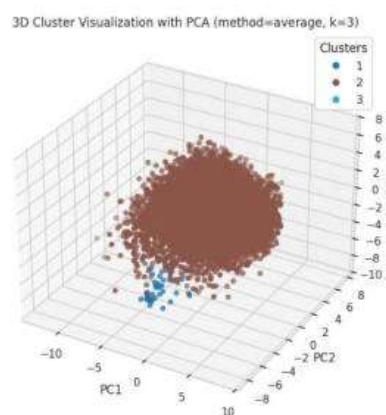
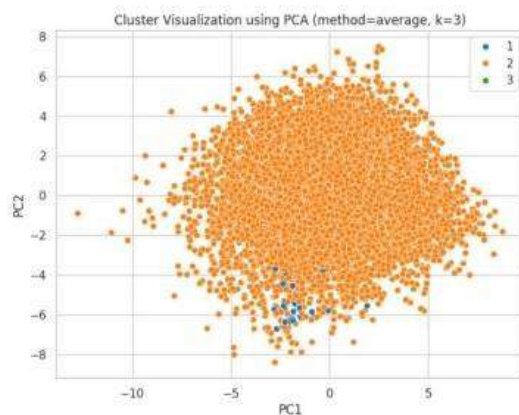
	Trait44	Trait45	Trait46	Trait47	Trait48	Trait49	Trait50
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Cluster

0	0.073395	0.100176	-0.035531	0.054071	0.009170	-0.565402	0.068147
1	-0.338241	0.513255	-0.353217	0.455935	0.190321	0.165903	0.502765
2	-0.200940	0.095203	-0.120820	0.260161	-0.055170	0.179234	0.046068
3	-0.210761	0.324626	-0.121280	0.353323	0.461777	0.153339	0.265931
4	-0.324666	0.302121	-0.332946	0.089306	0.353352	0.307455	0.449046
5	0.808724	-1.020074	0.877845	-0.868069	-0.662984	-0.502061	-1.053430
6	-0.220088	-0.050200	-0.265607	-0.034065	0.269410	0.428106	0.152469
7	0.517798	-0.177179	0.418057	-0.167483	-0.458883	-0.215691	-0.311002
8	-0.627458	-2.540539	-0.645258	-2.947603	-1.623423	-2.892337	-2.568443

[9 rows x 50 columns]

Hierarchical Clustering:



	Trait1	Trait2	Trait3	Trait4	Trait5	Trait6	Trait7 \
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Cluster

1	-0.557587	-0.910424	-1.452017	-1.510820	-1.488134	-1.074768	-1.222825
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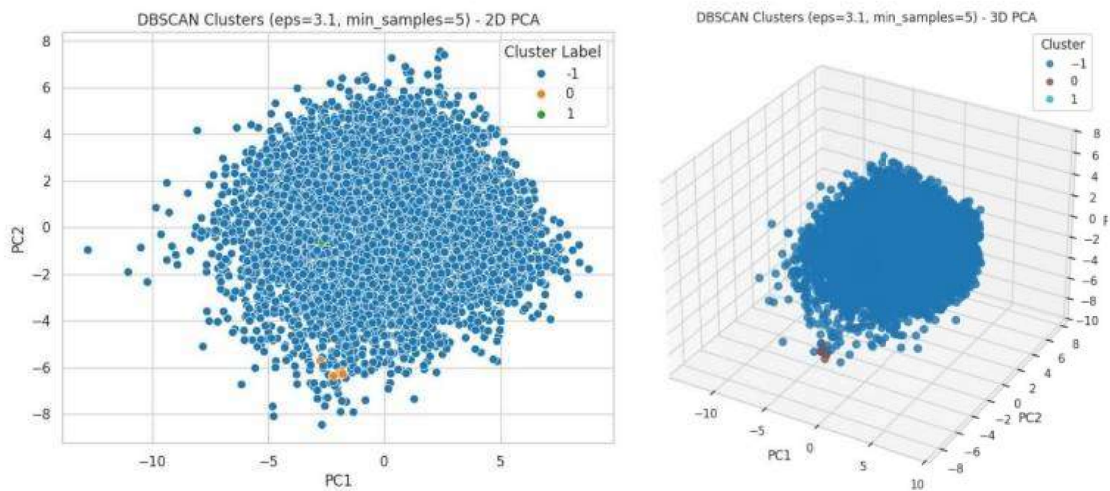
2	0.005526	0.003058	0.002592	0.010282	-0.006969	0.007328	-0.009896
3	1.866480	1.690583	-1.909883	1.514905	-1.819753	-0.334811	1.600671

	Trait8	Trait9	Trait10	...	Trait41	Trait42	Trait43 \
Cluster			...				
1	-1.859135	-1.377598	-1.778985	...	-1.837839	-0.587053	-2.368085
2	0.000625	0.012602	-0.001263	...	0.002971	0.003215	0.005279
3	1.255654	-1.477577	1.113565	...	-2.353117	2.642980	-2.803004

	Trait44	Trait45	Trait46	Trait47	Trait48	Trait49	Trait50
Cluster							
1	-0.668652	-2.517367	-0.687970	-2.979967	-1.674833	-2.905535	-2.611830
2	0.006528	0.010749	0.002077	0.017412	0.005385	0.013119	0.019100
3	2.769154	-1.862338	-0.825098	-3.048045	-1.769047	0.856872	-1.937849

[3 rows x 50 columns]

DBSCAN:



	Trait1	Trait2	Trait3	Trait4	Trait5	Trait6	Trait7 \
Cluster							
1	-0.556829	-0.910195	-1.445631	-1.511968	-1.472056	-1.071010	-1.204366
2	0.003175	0.005324	0.008919	0.008975	0.009068	0.006499	0.007108
3	1.853858	1.696612	-1.903478	1.513440	-1.803665	-0.334189	1.615742

	Trait8	Trait9	Trait10	...	Trait41	Trait42	Trait43 \
Cluster				...			
1	-1.850732	-1.390059	-1.761514	...	-1.818682	-0.584894	-2.347887
2	0.011045	0.008542	0.010521	...	0.011214	0.003266	0.014454
3	1.267436	-1.490642	1.121783	...	-2.330905	2.633603	-2.780785

	Trait44	Trait45	Trait46	Trait47	Trait48	Trait49	Trait50
Cluster							
1	-0.671203	-2.509525	-0.688216	-2.945533	-1.665025	-2.898732	-2.637202
2	0.003774	0.015336	0.004238	0.018085	0.010228	0.017413	0.016117
3	2.765568	-1.855339	-0.825823	-3.012844	-1.758961	0.860440	-1.957485

[3 rows x 50 columns]

We interpret clusters by examining patterns in their average trait scores:

KMeans (k=9)

Cluster 1: Has high positive values in many traits (e.g., Trait3, Trait5, Trait7, Trait10) and negative values in Trait6 and Trait8 — could represent individuals with generally high scores in positive traits but some emotional or behavioral instability.

Cluster 8: Extremely low scores across all traits (many <-1.5), possibly representing outliers, anomalous responses, or extreme personality types.

Clusters 2, 3, 5, 6: Mostly moderate scores, with some variation (e.g., Cluster 3 has high Trait4 & Trait5 but low Trait1, Trait3).

Cluster 7: Appears near-neutral on many traits — might represent a “balanced” or “average” group.

Hierarchical (average linkage, k=3)

Cluster 1: Very negative scores across nearly all traits — indicates a clearly distinct group with possibly low openness, agreeableness, and emotional stability.

Cluster 2: Very close to 0 across all traits — likely represents the average or central population.

Cluster 3: Has a mixture of extreme highs and lows (e.g., high Trait1, Trait2, Trait8; very low Trait3, Trait44) — indicating a unique, polarized personality pattern.

DBSCAN (eps=3.1, min_samples=5)

Cluster 1: Similar to Hierarchical's Cluster 1 — very low scores, again suggesting outliers or highly distinct negative traits.

Cluster 2: Very neutral values (~0) — likely the main population.

Cluster 3: Similar to Hierarchical Cluster 3 — high Trait1/2/8, low Trait3/44 — possibly an extroverted but emotionally unstable profile.

Conclusion: Trait patterns in DBSCAN and Hierarchical clusters closely align, suggesting consistency and stronger interpretability compared to KMeans.

KMeans formed too many small or similar clusters → low silhouette, weak separation.

Hierarchical + DBSCAN both identified 3 meaningful clusters with:

A clearly negative/extreme group (Cluster 1),

A neutral/average group (Cluster 2),

A high-polarity group (Cluster 3).

DBSCAN is especially useful here due to its explicit noise handling (label = -1), which KMeans/Hierarchical lack.

Final Recommendation

Best overall algorithm: Hierarchical Clustering (average linkage, k=3)

High silhouette score (~0.29),

Clear, interpretable clusters,

Matches DBSCAN's insights without sensitivity to parameter tuning.

i feel like if i chose more samples , for example instead of 10000 maybe 100000 the number of clusters could make more sense.

increasing the number of samples from 10,000 to 100,000 is likely to make clusters more meaningful, stable, and interpretable. It can justify the use of a larger number of clusters particularly for KMeans and can enhance the reliability of density-based clustering like DBSCAN. With more data, the clustering structure becomes clearer leading to deeper insights and more confident interpretations.

Conclusion

I In summary, this personality profiling exercise was successful in using a number of clustering algorithms on the Big Five Personality Test dataset and extracting significant patterns of personal personality traits. Out of the methods attempted, Hierarchical Clustering with average linkage (k=3) was the best, yielding intelligible and well-separated clusters that had the most similar results to DBSCAN's. These clusters illustrated distinct personality types: a neutral/mean group, an extreme or negative group, and a polarized group with wide variability between characteristics. While KMeans was beset by low silhouette scores and over-fragmentation, DBSCAN yielded valuable insight with noise detection and density-based clustering. The research also suggested the necessity for scaling, dimension reduction, and careful choice of metric. Finally, it was noted that when the sample size is increased to 100,000 from 10,000, then more solid and more detailed clustering results will be achieved, which can produce more insightful psychological observations and substantiate better clusters in models like KMeans.

References

<https://www.kaggle.com/datasets/tunguz/big-five-personality-test/data>

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